Appendix of DemandClean: Task-Driven Operation Selection for Data Cleaning with Constrained Ground-Truth

This appendix accompanies the paper. Numbers with the prefix A refer to this appendix, and all other section, theorem, table and figure numbers refer to the paper.

A Proof of Theorem 1

Ψ as a polynomial: Fix an execution order of the candidates. A deterministic policy is a vector $o = (o_1, \dots, o_K) \in \mathcal{O}^K$, the environment executes it in order and turns a deletion into a no-op once only $\lceil \rho n \rceil$ rows remain. Write $f(o) = \Phi(o) \in \mathbb{R}^2$ for the resulting state. Under π_p the vector o has probability $\prod_{i=1}^K p_{o_i}$, hence

$$\Psi(p) = \sum_{o \in \mathcal{O}^K} f(o) \prod_{i=1}^K p_{o_i},$$

a polynomial in p and therefore continuous on the compact set Δ. At the vertex $p = \mathbf{1}_o$ all mass sits on the policy that applies o to every candidate, so $\Psi(\mathbf{1}_o)$ is the state of the corresponding extreme strategy, the retention floor included for DeleteAll, and without a budget RepairAll repairs every candidate. The continuous image of the connected compact set Δ is connected and compact, and the images of the six edges of Δ are curves joining the anchors.

Bounded differences: Let o and o' differ only in coordinate i. The two executions differ in the value of the cell e_i, in the kept status of at most two rows, the row of e_i and the row of the first later deletion that the retention floor admits in one execution only, and in the VE-replace estimates of at most ξ later candidates, which VEC computes from the current table. A VE-replaced candidate counts as resolved whatever value it receives, so these estimates leave the resolved set unchanged. Let U and U' be the numbers of no-op candidates in kept rows and n_r, n_r' the numbers of kept rows. Then $|U - U'| \le 2\bar{k}$, $|n_r - n_r'| \le 2$, $n_r, n_r' \ge \rho n$ and $U \le K$, so

$$|A(o) - A(o')| \le \frac{|U - U'|}{n_r' d} + \frac{U}{d}\Big|\frac{1}{n_r} - \frac{1}{n_r'}\Big| \le \frac{2\bar{k}}{\rho n d} + \frac{2K}{\rho^2 n^2 d} \le \frac{1}{\rho n}\Big(\frac{2\bar{k}}{d} + \frac{2K}{\rho n d}\Big) =: c_A.$$

For diversity, $V = \frac{1}{d}\sum_c \frac{n_r}{n}\frac{\sigma_c^2(D')}{\sigma_c^2(D)}$. Changing one value or removing one row of a sample of size $m \ge \rho n$ with entries in $[-b, b]$ changes its sample variance by at most $\frac{8b^2}{m-1}$, and the factor $\frac{n_r}{n}$ changes by at most $\frac{1}{n}$. The changed cell, the at most ξ re-estimated cells and the at most two rows give at most $3 + \xi$ such changes. Hence

$$|V(o) - V(o')| \le \frac{3 + \xi}{s_{\min}^2}\Big(\frac{8b^2}{\rho n - 1} + \frac{2b^2}{n}\Big) =: c_V.$$

Concentration: The coordinates of o are independent under π_p. McDiarmid's inequality applied to A and to V gives, for $t > 0$, $\Pr[|A(\pi_p) - \Psi_A(p)| \ge t] \le 2\exp(-\frac{2t^2}{K c_A^2})$ and the same with c_V. Choosing $t = \max(c_A, c_V)\sqrt{\frac{K}{2}\log\frac{4}{\delta}}$ and a union bound over the two coordinates give $\|\Phi(\pi_p) - \Psi(p)\|_\infty \le t$ with probability at least $1 - \delta$. Since $\rho n \ge 2$ gives $\rho n - 1 \ge \frac{\rho n}{2}$ and each row holds at most $\bar{k}$ candidates, $n \ge \frac{K}{\bar{k}}$ and $\max(c_A, c_V) \le \frac{C'}{n} \le \frac{C'\bar{k}}{K}$ with C' depending on $\bar{k}, \xi, \rho, b, s_{\min}^2$ and $\frac{K}{nd}$, and $t \le C\sqrt{\frac{1}{K}\log\frac{4}{\delta}}$ with $C = \frac{C'\bar{k}}{\sqrt{2}}$. An event of positive probability has a realization, which is the deterministic policy of the statement. No step uses additivity of A or V over cells. The row-level coupling created by deletion enters only through $\bar{k}$ and ρ.

Directions: In $(\Delta A, \Delta R_{sam}, \Delta R_{var}, cost)$ no-op gives $(0, 0, 0, 0)$. op_{gt} and op_{ve} resolve one candidate without removing a row, which raises A by $\frac{1}{n_r d}$, leaves R_{sam} unchanged and changes one column variance by $O(\frac{1}{n})$, and only op_{gt} consumes budget. An admitted deletion removes the candidate's row, which multiplies the sample-retention factor $R_{sam} = \frac{n_r}{n}$ of every column by $\frac{n_r - 1}{n_r}$, so it is the only operation with $\Delta R_{sam} \ne 0$, and it raises $A = 1 - \frac{U}{n_r d}$ exactly when the removed row holds more than $\frac{U}{n_r}$ unresolved candidates. op_{gt} writes the true value by definition, and op_{ve} writes it only when VEC recovers it. □

B Proof of Theorem 2

Membership: A certificate is a policy $\pi : E \to \mathcal{O}$. Its cost is a sum of K terms, and $Perf(M; D_\pi)$ needs one application of π, one training run and one evaluation. The instance lists the GT value of every candidate, every level

of Algorithm A1 scans the table once per rule and its DC level returns a constant of the violated clause, so applying π takes polynomial time, and training and evaluation are polynomial by the condition on M, with a fixed seed for a randomized learner.

Per-cell costs: Per-cell costs contain the unit-cost case below, so hardness under per-cell costs follows from it.

Unit costs: Let $G = (\mathcal{U}, \mathcal{F})$ and k_G be a VERTEX COVER instance with $m = |\mathcal{F}|$. The instance gives the encoded features directly, and the classifier and VE-replace measure Euclidean distance on them. Work in $\mathbb{R}^{N_0}$ with one coordinate per vertex, one per edge and private coordinates for auxiliary rows, and fix $h = 8$ and $L = 7$, so that $r = \sqrt{\frac{h^2}{2} + L^2} = 9$, together with a rational $\tau \in (0, 1)$, $\eta = \frac{\tau^2}{8r}$ and a rational $\epsilon < \eta$, so that every coordinate is rational. The rule library is empty. All labels are binary and the true label of every validation point is 1.

- *Vertex rows.* For $u \in \mathcal{U}$, a training row $t_u = h\mathbf{1}_u$ with the unit vector $\mathbf{1}_u$ with true label 1 and observed label 0. Its label cell is the only candidate of the row and costs 1.
- *Validation points.* For $(u, w) \in \mathcal{F}$, a point $v_{uw} = \frac{h}{2}(\mathbf{1}_u + \mathbf{1}_w) + Lf_{uw}$ with a private unit vector f_{uw}. Then $\|v_{uw} - t_u\| = \|v_{uw} - t_w\| = r$, while every other vertex row lies at distance $\sqrt{\frac{3h^2}{2} + L^2} > r$ and every anchor or blocker built for another edge lies at distance at least $\sqrt{2}L - \epsilon > r + 3\eta$ from v_{uw}.
- *Anchors and blockers.* Clean rows $a_{uw} = v_{uw} + \epsilon g_{uw}$ with a private unit vector g_{uw} and label 1, and clean rows β_{uw}, β'_{uw} with label 0 placed on private axes at distances $r + \eta$ and $r + 2\eta$ from v_{uw}. These rows are not candidates, so no operation can touch them.
- *Decoys.* Five clean rows with label 0 at distance τ from each t_u along private axes. From v_{uw} they lie at distance $\sqrt{r^2 + \tau^2} > r + 2\eta$, and every other training row lies at distance at least $r > \tau$ from t_u, so the decoys are the five nearest kept rows of t_u.

The three nearest training rows of v_{uw} are a_{uw}, t_u, t_w, so with the observed labels $(1, 0, 0)$ the point is misclassified and it becomes correct exactly when t_u or t_w is repaired. Deleting t_u brings β_{uw} into the neighborhood, giving $(1, \ell_w, 0)$, which is correct only if t_w is repaired, so deletion never helps, and deleting both endpoints gives $(1, 0, 0)$. VE-replace on a label cell writes the majority label of the five nearest kept rows (Section E), which are the five decoys, so op_{ve} writes label 0 and equals op_{noop}. Consequently the largest number of correctly classified validation points under k_G unit-cost repairs equals the largest number of edges covered by k_G vertices, and accuracy 1 (that is, $\theta = 1$) is reachable with budget $B = k_G$ if and only if G has a vertex cover of size at most k_G. The instance has $|\mathcal{U}|(1 + 5) + 3m$ training rows and m validation points with polynomially bounded coordinates, so the reduction is polynomial, and VERTEX COVER is NP-complete in the strong sense. □

C SUFFICIENCY OF THE QUALITY STATE

Proposition 1 bounds the loss of planning in (A, V) by the fiber oscillation ω, and the diversity V of Definition 6 records sample size and per-column variance only. This section measures ω on the nine benchmark datasets and tests whether three statistics that (A, V) does not encode, the correlation structure, the class distribution and the shape of the joint distribution, predict the downstream score beyond (A, V).

Setup. On each dataset we draw 600 random policies over the four operations of Definition 9. A policy draws operation probabilities $p \sim \text{Dirichlet}(1, 1, 1, 1)$, deletes each training row that holds an oracle candidate with probability p_{del}, and resolves each candidate of the kept rows by no-op, GT-repair or VE-replace in proportion to the other three probabilities, with the value of REPLACEALL as the VE estimate. For every output D' with encoding X' we record A and V, where V averages over the columns that vary in the dirty training split, the correlation distance $\frac{\|C(X') - C(X_{clean})\|_F}{\|C(X_{dirty}) - C(X_{clean})\|_F}$ with C the column correlation matrix, the total-variation distance between the label distributions of D' and of the clean training split, the sliced Wasserstein distance between X' and X_{clean}, and the score of the first-choice model, clipped to $[0, 1]$ and computed as in Table IV. On `har` the silhouette is computed on a fixed subsample of 10,000 kept rows. The output of DEMANDCLEAN is measured the same way. We cut the (A, V) range of the random policies into a 20×20 grid and take $\omega(q)$ as the score range of the policies in a cell q that holds at least three of them.

Fiber oscillation. Table A1 reports ω next to the gain G of the clean data over NOFIX. The median oscillation is smaller than G on all nine datasets and at most 0.073, so policies that share a quality state usually score alike. The tail is heavier on `nasa`, `soilmoisture` and `mercedes`, where the largest oscillation within a cell reaches 0.535, 0.843 and 0.379. On these datasets the bound $2 \sup_q \omega(q)$ of Proposition 1 is loose, which is the case that the local features of the 10-D state address (Section IV-B2).

Statistics beyond (A, V). Adding the correlation distance, the label-distribution distance and the sliced Wasserstein distance to (A, V) raises the cross-validated R^2 of the score by at most 0.03 on six of the nine datasets. The increase is 0.07 on `soilmoisture` and 0.18 on `beers` and `har`, so on these datasets correlation structure and distribution shape carry information beyond (A, V). On `beers` and `har` the output of DEMANDCLEAN is closer to the clean training split than at least 99% of the random policies in both correlation structure and distribution shape, and on `beers` it keeps the label distribution of NOFIX. On `mercedes` and `smartfactory` the output of DEMANDCLEAN is farther from the clean correlation structure than most random policies and still outscores all of them, the regime of Section V-C1 in which retained variance outweighs agreement with the clean data.

Where DEMANDCLEAN lands. On the eight datasets other than `adult`, DEMANDCLEAN outscores at least 90% of the 20 random policies nearest to it in the standardized (A, V) plane and at least 91% of all random policies. GT does not account for this, since among the random policies that read at least as many GT cells as DEMANDCLEAN, it outscores at least 85% on each of the seven datasets where such policies exist. Within a cell of Q, the policy of DEMANDCLEAN thus lands at the high end of the fiber, the part of the planning

TABLE A1
Sufficiency of the quality state on the nine benchmark datasets. G: gain of the clean data over NoFix. ω_{50}, ω_{90}: median and 90th percentile of the fiber oscillation over the grid cells. R^2_{AV}, R^2_+: five-fold cross-validated R^2 of a 10-nearest-neighbor regressor that predicts the score from (A, V), and from (A, V) with the three further statistics. Near, All: share (%) of the 20 random policies nearest to DemandClean in (A, V), and of all 600 random policies, that score below DemandClean.

Dataset	G	ω_{50}	ω_{90}	R^2_{AV}	R^2_+	Near	All
adult	.005	.003	.004	.10	.09	15	26
beers	.040	.029	.044	.39	.57	90	99
breast_c.	.029	.014	.021	.33	.34	95	98
smartf.	.005	.004	.007	.77	.80	100	100
bike	.141	.011	.020	.96	.96	100	100
mercedes	.034	.012	.023	.43	.43	100	100
nasa	.532	.000	.177	.81	.83	90	100
soilm.	.102	.045	.082	.47	.55	100	91
har	.408	.073	.151	.78	.97	100	100

error that Proposition 1 leaves open. On adult the scores of all random policies span less than 0.01, DemandClean spends only the calibration seed set, and its score lies 0.005 below the best random policy.

D Method Details

Detection example. Figure 2 (Step 1 Example) illustrates the four stages on a Beer subset. In Stage 1, the IBU cell is flagged as missing (null), but insufficient information prevents repair at this stage. In Stage 2, ABV = 178.7 is flagged by the domain rule ABV<100 and provisionally corrected to 45, the most frequent ABV value of the column. With the ABV column stabilized, Stage 3 evaluates the CFD ABV<50 $\Rightarrow$ City=Gary reliably and flags the in-domain City=Chicago as a semantic error.

Inference modes. The trained agent operates in pure exploitation mode ($\varepsilon = 0$), processing each detected error sequentially via the two-stage decision (Section IV-B2), as the Two-Phase Inference panel of Figure 2 shows. Phase 1 (agent decision) issues one operation per error cell, and the zero-cost actions (op_{noop}, op_{del}, op_{ve}) execute immediately. A GT-repair (op_{gt}) is realized in one of two modes (Algorithm 3). In the online mode, labels are available at decision time, so each GT-repair writes its true value on the spot and the later states are exact. In the offline mode, every cell assigned GT-repair is recorded into the *repair plan* $P = \{(row_i, col_i, v_i)\}_{i=1}^{|P|}$, with its VEC estimate v_i temporarily written to preserve state continuity. The plan is thus generated without touching any GT (① in Figure 2), and the annotation demand drops from $|E|$ to $|P|$ ($|P| \ll |E|$). Phase 2 (action execution) then writes the GT that users provide for any subset of P (②), so users can batch and defer the labeling of P.

Policies within one cell. Two policies in one cell differ in performance in two cases. They may resolve the same candidates by different operations, since exchanging GT-repair and VE-replace leaves A unchanged (Theorem 3(i)) while only GT-repair always restores the true value, or they may delete equally many but different rows, which changes the training samples that the model sees. The quality state alone cannot separate the policies of one cell. In the 10-D state (Section IV-B2), the kept-row ratio (s_6), the variance ratio of the current column (s_7) and the remaining-error ratio (s_9) track the episode's progress and the surrounding quality statistics, and the local features and budget awareness separate the policies inside it, so that the agent can move toward the best cell under sequential per-cell decisions. These features are computed identically regardless of a dataset's domain, feature count, or error pattern, so the same agent runs unchanged on another dataset (Section V-D). Because the state is computed per cell, an agent selects different operations on datasets whose candidates differ, which is consistent with the dataset-dependent attribution of Section V-C3. For example, when the remaining budget ratio (s_8) falls below the remaining error ratio (s_9), the agent prefers zero-cost VE-replace to GT-repair.

E Implementation Details

Data split. Every dataset is split into training, validation and test rows in the proportions 60%, 20% and 20%. The baselines and every reported DemandClean run use this split. Categorical features are integer-encoded and numerical features are standardized, with encoders fitted on the dirty training rows.

Candidate downstream models. The tolerance-based selection of Section IV-B1 chooses among random forest (RF), logistic regression (LR), SVM, k-nearest neighbors (KNN), decision tree (DT) and gradient boosting (GB) for classification, among RF, linear regression (LR), Ridge, Lasso, KNN and GB for regression, and among KMeans, agglomerative and spectral clustering for clustering.

Value estimation chain. Algorithm A1 checks the estimation levels in a fixed priority order and adopts the first level that returns a value. Every level returns its value through the projection clip_j, which maps a value onto the declared [DOMAIN] range of an integer or real column j and leaves other columns unchanged. VE-replace on a label cell writes the majority label of the 5 nearest kept rows for classification and their inverse-distance-weighted mean for regression.

State. Table A2 gives the ten coordinates for a candidate in row i and column j of a table with n rows and d feature columns, where E is the candidate set of the episode, k the position of the current candidate in the processing order, n_r the number of kept rows, B the GT budget in cost units and C the cost spent so far. The proxy model of the reward supplies s_1 and s_2. A candidate in a deleted row receives zeros in s_0 to s_7.

Reward. The proxy score sc is the accuracy for classification, $\frac{1}{1+\ln(1+\mathrm{MSE})}$ for regression and $\frac{1+sil}{2}$ for clustering, computed on the validation split. sc_0 is the score of the proxy fitted on the rows of the episode table that hold no candidate, and sc_k the score after the proxy is refitted on the kept rows at step k. With $\pi_k = \frac{(1-s_2)+s_1+s_5}{3}$ and $g_k = \frac{1}{1+e^{-6(s_8-s_9)}}$, the shaping term of action a_k is

$$r_k^{sh} = \kappa\, w(t) \cdot \begin{cases} -0.03 & a_k = op_{noop}, \\ (0.02 + 0.04\,\pi_k)\, g_k & a_k = op_{gt}, \\ \delta_{del} & a_k = op_{del}, \\ 0.03 & a_k = op_{ve}, \end{cases}$$

Algorithm A1 Value Estimation Chain for a feature cell (i, j)

Require: Current table X, deleted rows R_{del}, column means μ, rule library $\mathcal{R}$, raw dirty string $\tilde{x}_{ij}$
Ensure: Estimate $\hat{v}$

// L1 FD
for each FD $L \Rightarrow j$ in $\mathcal{R}$ with $X_{i,L}$ complete **do**
  $S \leftarrow \{i' \neq i,\ i' \notin R_{del} : X_{i',L} = X_{i,L},\ X_{i'j}$ present$\}$
  if $|S| \geq 2$ **return** clip_j(majority of $X_{S,j}$ if j is categorical or $X_{S,j}$ has at most 5 distinct values, else mean of $X_{S,j}$)
end for
// L2 DC
for each DC that marks j and whose clauses outside j hold on row i **do**
  if its clause on j is $x_j \neq a$, $x_j > a$, $x_j \geq a$, $x_j < a$ or $x_j \leq a$ **return** $\mathrm{clip}_j(a)$
  if its clause is $|x_j - x_l| > t$ **return** $\mathrm{clip}_j(X_{il})$
end for
// L3 CFD
if a CFD with target j declares a baseline b **return** $\mathrm{clip}_j(b)$
// L4 numeric extraction
if j is numerical, X_{ij} is missing and $\tilde{x}_{ij}$ contains a match of `-?\d+(\.\d+)?` **return** clip_j(first match)
// L5 edit distance
$c^\star \leftarrow \arg\max_c \mathrm{sim}(\tilde{x}_{ij}, c)$ over the known categories of j
if j is categorical, $\mathrm{sim}(\tilde{x}_{ij}, c^\star) \geq 0.6$, $c^\star \neq \tilde{x}_{ij}$ and X_{ij} is not the mode of j **return** $c^\star$
// L6 nearest neighbors
$N \leftarrow$ the 5 rows $i' \notin R_{del} \cup \{i\}$ with $X_{i'j}$ present that are nearest to row i in Euclidean distance over the other columns, missing entries filled by μ
$w_{i'} \propto \frac{1}{dist(i,i')+10^{-8}}$
if $N \neq \emptyset$ **return** clip_j(weighted vote of $X_{N,j}$ if j is categorical, else $\sum_{i' \in N} w_{i'} X_{i'j}$)
// L7 column statistic
return clip_j(mode of j if j is categorical, else μ_j)

TABLE A2
The 10-dimensional state of a candidate cell (i, j). Label cells (j is the label column) use the values in the last column.

	Feature cell	Label cell
s_0	$\frac{t}{3}$ for type $t \in \{0, 1, 2, 3\}$ (missing, semantic, syntactic, label)	1
s_1	importance of column j in the proxy, normalized to sum one (RF impurity importance, absolute Ridge coefficients, mean centroid deviation for KMeans)	1 (max. for KMeans)
s_2	classification $2 \max_c \hat{p}(c \mid x_i) - 1$, regression $\frac{\lvert\hat{y}_i - \bar{y}\rvert}{2\sigma_y}$, clustering $\delta_2 - \delta_1$, with δ_1, δ_2 the distances to the two nearest centroids, all clipped to $[0, 1]$	same
s_3	$\frac{i}{n-1}$	$\frac{i}{n-1}$
s_4	$\frac{j}{d-1}$	1
s_5	$\frac{\lvert E_j\rvert}{\lvert E\rvert}$, the share of candidates in column j	0.5
s_6	$R_{sam} = \frac{n_r}{n}$	$\frac{n_r}{n}$
s_7	$R_{var} = \min\big(1.5, \frac{\sigma_j^2\,\text{(kept rows)}}{\sigma_j^2\,\text{(episode start)}}\big)$	1
s_8	$\frac{\max(0, B-C)}{\max(B,1)}$	same
s_9	$\frac{\lvert E\rvert - k}{\lvert E\rvert}$	same

where $\kappa = \min\big(10, \max(0.1, \frac{m}{0.03})\big)$, m is the exponential average of $|sc_k - sc_0|$ over the evaluation steps of the episode, set to the first value and then updated by $m \leftarrow 0.3\,|sc_k - sc_0| + 0.7\,m$ ($\kappa = 1$ before the first evaluation step), and the shaping weight is $w(t) = 1$ in the first half of the T episodes and $w(t) = \max\big(0.1,\ 1 - 0.9\,\frac{2t-T}{T}\big)$ afterwards. Every N-th step, and the first one, is an evaluation step with reward $sc_k - sc_0 - \lambda c_k \mathbf{1}[a_k = op_{gt}] + r_k^{sh}$, the other steps receive r_k^{sh}, and

$$N = \max\Big(5,\ \min\big(\max(1, \lfloor\tfrac{|E|\,n}{5\cdot 10^5}\rfloor),\ \max(1, \lfloor\tfrac{|E|}{20}\rfloor),\ 200\big)\Big).$$

The last step of the episode receives $R_{final} = 5(sc_T - sc_0) + \beta\big(1 - \frac{|R_{del}|}{n}\big) - \lambda C$ in place of the step reward. The budget is $B = \lfloor r_{\max}|E|\rfloor$ cost units, a GT-repair request with $C + c_k > B$ is executed as VE-replace, and a deletion is executed as no-op once $\lfloor 0.8\,n\rfloor$ rows are deleted. All reported runs use unit costs $c_k = 1$. The per-dataset constants are listed in our repository.

Network training. Each stage is a Dueling network with two hidden layers of 128 ReLU units and output $Q(s, a) = V(s) + A(s, a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s, a')$, trained with the Double DQN target, squared loss and Adam at learning rate $5\cdot 10^{-4}$, with discount 0.95 and a replay buffer of 10^4 transitions per stage. An episode is one pass over the shuffled candidates of one corrupted copy of D_{base}. After each episode both stages take one update on a mini-batch of 64, the exploration rate is multiplied by 0.995 from 1.0 down to 0.1, and every fifth episode the target networks move by $\theta^- \leftarrow 0.1\,\theta + 0.9\,\theta^-$. Stage 1 stores every transition, and Stage 2 stores the transitions on which a cell was resolved in place by GT-repair or VE-replace. Training keeps the weights of the episode with the highest final proxy score, ties going to the episode with fewer GT repairs, and restores them at the end. D_{base} is the better of two zero-cost tables under the 5-fold cross-validated proxy score multiplied by $\sqrt{\frac{n_{kept}}{n}}$. The first deletes the rows that hold detected candidates while keeping at least $\max(10, \lfloor 0.2n\rfloor)$ rows and fills the remaining missing cells by the value estimation chain, and the second applies the value estimation chain to every detected cell and the 5-nearest-neighbor label estimate to every detected label.

Adversarial error injection. For each error type the detector output fixes a rate $\hat{r}$, the number of detected cells of that type divided by nd for missing, semantic and syntactic errors and by n for label errors. Each episode draws its rate uniformly from $[\ell, u]$ with $\ell = \max(0.8\hat{r}, 0.005)$ and $u = \max(1.2\hat{r}, \ell + 0.005)$, uses $[0, 0]$ for a type without detections, and uses $[0.005, 0.02]$ for semantic errors when no semantic cell is detected but the library holds FDs, CFDs or DCs. With n_b rows in D_{base}, the injector attempts $\lfloor n_b r\rfloor$ cells per type at random positions not used before, and the value of each cell before injection is its pseudo-GT. Missing errors blank the cell. When the library holds CFDs or DCs, half of the semantic quota goes to CFD violations, which move the target value past the CFD threshold by $U(0, 2)$ raw units while staying inside its declared domain, and the rest goes to DC violations that make every clause of a DC hold. The quota that these cannot place goes to FD violations, which copy the right-hand value of another left-hand group into at most $\lfloor\frac{g-1}{2}\rfloor$ rows of a group of $g \geq 3$ rows, so that the original value keeps the majority, and cells that no rule can place receive another value of their column. Syntactic errors give 30% of their quota to domain violations when the library declares domains, which place a numeric value 1 to 5 raw units outside its range or an enumerated value 1 to 3 steps beyond the largest one. The remainder is split 40%, 30% and 30% among Gaussian shifts to the column mean plus or minus $U(3, 5)$ standard deviations, values $U(0.3, 1.0)$ times the 1% to 99% quantile range beyond one

of those quantiles, and magnitude errors that multiply the value by 10 or by 11 after rounding, as in $3 \rightarrow 33$, or move it across zero. These three draw from columns outside the left-hand sides of FDs, and on a categorical column they substitute another valid category. Label errors are injected only when the detector reports label errors, by flips that follow the detected flip pattern for classification and by Gaussian noise of the detected spread for regression, in rows that carry no injected feature error.

Detection. The calibration seed set of the ensemble detector consists of 20 to 80 labeled records on the nine benchmark datasets, and all of their 120 to 7,540 cells count toward GT%. The runs on `flights`, `soccer` and `hospital` disable this detector and read no labeled cell for detection.

Baselines. The baselines operate on the training rows of the split described above. DAGPIPE also transforms the test inputs with the graph learned on these rows, keeping its structure and operator parameters fixed. The common evaluator trains the downstream model of Table III on every output and scores it on the test rows, except for clustering, whose silhouette is computed on the cleaned training rows. GT counts the cells of the clean table that a method reads on the training rows. Our repository (https://github.com/qzkinhit/DemandClean) documents the implementation of every baseline, its use of GT and of the validation split, and the adaptations that the regression, clustering and organically dirty datasets require. A method that outputs a cleaned matrix in the encoding of the evaluator, which is the case for DIFFPREP, DIANA and LEARN2CLEAN, is scored on that matrix, and DAGPIPE is scored on the features that its graph produces.

Rule libraries. DEMANDCLEAN reads the library $\mathcal{R}$ in `data/<ds>/rules.txt`, whose `[DOMAIN]`, `[FD]`, `[CFD]`, `[DC]` and `[REGEX]` sections feed its detector, its value estimation chain and its injector. HOLOCLEAN, UNICLEAN and HORIZON read the rule file of the Clean4ML benchmark release, `benchmark/Data/<ds>/rules.txt`, whose `[HOLOCLEAN_DC]`, `[UNICLEAN]` and `[HORIZON_FD]` sections state the dependencies in the language of each method. On the three organic datasets they read the denial constraints, cleaners and dependencies released with UNICLEAN. Table A3 counts the rules of both files. Both files hold the same functional dependencies on `adult`, `beers`, `har` and `nasa`. On `bike`, $\mathcal{R}$ states the relation between season and month by six single-tuple denial constraints and relates holiday and weekday to workingday by three more, while the baselines read `mnth` $\Rightarrow$ `season`.

The `[DOMAIN]` sections of $\mathcal{R}$ list, for each constrained column, the range or the value set that the column takes. The numeric bounds of `soilmoisture` and of the date column of `bike` are rounded, and the `ounces` set of `beers` lists the container sizes. On the organic datasets the domains constrain the label column, and on `hospital` also two categorical columns. The experiment of Table A6 also deploys DEMANDCLEAN with libraries derived from the dirty table alone, whose numeric domains come from the 0.5% and 99.5% quantiles of the dirty column, whose categorical domains hold the values that occur at least twice, and whose FDs are mined from the dirty table.

TABLE A3
Rules per dataset. Left, the library $\mathcal{R}$ of DEMANDCLEAN by section. Right, the denial constraints of HOLOCLEAN, the cleaners of UNICLEAN and the FDs of HORIZON in the Clean4ML rule file.

	$\mathcal{R}$ of DEMANDCLEAN					Clean4ML file		
Dataset	Dom.	FD	CFD	DC	Regex	HC	UC	Hor
`adult`	6	1	24	0	0	1	6	1
`beers`	4	4	0	0	0	4	5	4
`bike`	16	0	0	9	0	1	8	1
`breast_c.`	10	0	22	8	3	1	9	1
`har`	4	1	0	0	0	1	6	1
`mercedes`	9	0	0	0	0	2	10	2
`nasa`	6	3	0	0	0	3	9	3
`smartf.`	19	0	0	0	0	0	5	12
`soilmoist.`	127	0	0	124	0	0	10	1

TABLE A4
Tol_{rel} of agents deployed without retraining across datasets of the same task type. Rows: dataset the agent was trained on. Columns: target dataset. Bold diagonal: the agent trained on the target.

Classification Source \ Target	`adult`	`beers`	`breast_c.`	`smartf.`
`adult`	**.31**	−.37	.25	−.45
`beers`	−.90	**1.00**	1.00	1.14
`breast_c.`	−.52	1.00	**1.00**	.73
`smartf.`	−.90	.53	1.00	**.82**
Regression Source \ Target	`bike`	`mercedes`	`nasa`	`soilm.`
`bike`	**1.00**	1.35	.08	.78
`mercedes`	.72	**.53**	.07	.67
`nasa`	1.00	1.26	**1.00**	.81
`soilm.`	.94	.88	.00	**.79**

F ADDITIONAL RESULTS

Transfer without retraining. An agent trained on one dataset is deployed without retraining on every other dataset of the same task type, with the detector of the target dataset, and Table A4 reports its Tol_{rel} on the target. Every entry, the diagonal included, redeploys a trained agent under this protocol, so each column compares the transferred agents with the agent of the target. On `beers`, `breast_cancer`, `smartfactory`, `bike`, `mercedes` and `soilmoisture`, at least one transferred agent matches or exceeds the agent trained on the target, and on `mercedes` every transferred agent exceeds it. On `adult` and `nasa` the agent trained on the target stays ahead of every transferred agent.

GT-consuming baselines. Table A5 runs ACTIVECLEAN, COMET, LOPSTER and RetClean at two GT levels against the deployed run of DEMANDCLEAN. At the budget of DEMANDCLEAN, ACTIVECLEAN cleans and LOPSTER learns from as many records as the GT cells of DEMANDCLEAN cover, and COMET may query as many GT values as DEMANDCLEAN has seed records and GT-repairs together. RetClean retrieves its repairs from a data lake of clean values, which at the lower level holds the larger of 1.65% of the cells and the GT cells that DEMANDCLEAN reads on the dataset, so it never holds less GT than DEMANDCLEAN. At the default level, ACTIVECLEAN and COMET use their default budgets, LOPSTER learns from half of the training rows, and the lake of RetClean holds the clean values

TABLE A5
Downstream score and GT% of the four GT-learning baselines at no less than the GT budget of DEMANDCLEAN and at their default GT use, scored as in Table IV.

	DEMANDCLEAN		ACTIVECLEAN				COMET				LOPSTER				RetClean			
			DEMANDCLEAN budget		default		DEMANDCLEAN budget		default		DEMANDCLEAN budget		default		≥ DEMANDCLEAN budget		default	
	Score	GT%	Score	GT%	Score	GT%	Score	GT%	Score	GT%	Score	GT%	Score	GT%	Score	GT%	Score	GT%
adult	.854	0.1	.851	0.1	.854	1.1	.852	56.0	.852	56.0	.805	0.1	.764	30.0	.852	0	.852	0
beers	.361	1.8	.326	1.7	.340	20.7	.340	48.0	.344	48.0	.111	1.7	.173	30.0	.303	1.8	.353	27.0
breast_c.	.993	2.9	.957	2.9	.986	29.9	.957	53.9	.979	53.9	.964	2.9	.943	29.9	.986	3.0	.986	29.9
smartf.	.969	0.2	.962	0.2	.963	2.1	.962	56.8	.966	56.8	.702	0.2	.642	30.0	.965	1.6	.962	30.0
bike	1.00	1.0	.895	1.0	.943	2.9	1.00	56.2	1.00	56.2	.955	1.0	.984	30.0	.983	1.6	.982	30.0
mercedes	.428	2.0	.394	2.0	.368	11.9	.360	59.8	.363	59.8	.401	2.0	.408	30.0	.360	2.0	.358	30.0
nasa	.533	6.5	−.001	6.5	.000	29.9	.532	50.0	.390	50.0	.164	6.5	.417	29.9	.509	6.5	.515	29.9
soilm.	.940	2.9	.872	2.9	.937	29.9	.857	59.5	.857	59.5	.868	2.9	.860	29.9	.904	2.9	.913	29.9
har	.899	0.1	.499	0.1	.506	0.7	.504	3.2	.520	3.2	.747	0.1	.866	30.0	.799	1.6	.802	30.0

TABLE A6
Detection F1 and Tol_{rel} under degraded rule libraries, averaged over the nine datasets. Each removal level and the false-constraint library average three random draws.

Rule library	Detection F1	Tol_{rel}
Full	.84	.83
25% / 50% / 75% removed	.84 / .82 / .80	.78 / .70 / .64
False constraints added	.46	.31
Derived from the dirty table	.72	.48

of a fixed random half of the training rows. The GT of COMET includes its error mask at both levels. On adult RetClean finds no target cell and reads no GT. flights, soccer and hospital are left out of the table, because DEMANDCLEAN reads at most four GT cells on them, and at this budget ACTIVECLEAN, COMET and LOPSTER score exactly as NOFIX. On these three datasets RetClean scores .983, 1.00 and .995 at the lower level and .998, 1.00 and .995 at the default level, against 1.00 for DEMANDCLEAN on each.

Robustness to the rule library. Every trained agent is redeployed with its full library and with libraries that lose a quarter to three quarters of their constraints, gain false dependencies and narrowed domains, or are derived from the dirty table alone, all under the same protocol (Table A6). Removing constraints degrades the result gradually. Adding false constraints harms the result most. Missed cells never enter the candidate set, so detection recall bounds the reachable gain.

GT usage per dataset. Table A8 lists the GT usage behind the GT% column of Table IV and the GT panel of Figure 4, counted by metric ⑤ of Section V-B. The oracle-detection references read every clean training cell, COMET reads the clean values of the training rows it processes through its error mask, and every baseline that the table does not list reads no clean value.

GT allocation. DEMANDCLEAN spends GT where the data side has room to improve. On adult the four reference strategies of Table IV differ by less than 0.005, the prior tolerance of the selected model is close to zero before any labeling, and DEMANDCLEAN accordingly spends no GT beyond the calibration seed set. On smartfactory repairing every error also gains less than 0.005 over NOFIX, and DEMANDCLEAN still exceeds the clean data with 0.2% of the cells as GT. On hospital the dirty data already reaches the top score, as do REPAIRALL and most baselines, and DEMANDCLEAN reaches it without any GT. On soilmoisture DEMANDCLEAN recovers most of the gain of the clean data with the calibration seed set as its only GT.

Organically dirty datasets. Table A10 lists the four organically dirty datasets of Table IV with the input tokens and the cost of the FM baseline. Over all twelve datasets the FM baseline costs $207.30 at list price, $98.27 of it on mercedes.

Runtime. Table A7 lists every reference strategy, the detection, the model selection and the wall-clock runtime of DEMANDCLEAN and of four baselines. Runtimes are measured on an Apple M2 SoC with 16 GB unified memory. Training dominates, ranging from 37 seconds (breast_cancer) to 6.7 hours (har). Each episode makes $O(|E|)$ forward passes through two shallow MLPs and a roughly constant number of proxy refits, and needs no GPU. har (70K rows) takes 6.7 hours, with each evaluation-step reward refitting the proxy on all of its rows, whereas mercedes (4K rows, 181K error cells) finishes in 1.6 hours, with Ridge refitting touching only 2.5K rows. Inference ranges from under 1 second (nasa) to 200 seconds (bike) and is shorter than the end-to-end run of RAHA+BARAN, HOLOCLEAN and UNICLEAN on eight of nine datasets. Training runs once per dataset, every later cleaning run on that dataset costs only this inference time.

Variant 1: DEMANDCLEAN-ND (oracle detection). Replacing the detector of Section IV-A with oracle error cells lowers Tol_{rel} on all eight datasets, on soilmoisture below NOFIX. On mercedes and smartfactory the detector is already near exact ($P_d \geq 0.98$, $R_d = 1.00$), so the loss cannot come from detection quality. What changes is the typed error set the agent receives, because the oracle path bypasses the staged typing and provisional VEC repairs of Algorithm 1, shifting the error type (s_0) and the column error rate (s_5) in the state even where the flagged cells coincide. Full DEMANDCLEAN is thus both stronger and steadier than the variant. Its mean of 1.22 exceeds the clean data against 0.61 for the variant, and its interquartile range over the eight datasets is 0.32, less than half the 0.76 of the variant.

Variant 2: DEMANDCLEAN-NGT (no GT). The zero-budget agent retains performance on exactly the datasets where VE-replace carries the gain (Finding 3), and drops to the NOFIX floor or below where precise repair or deletion is needed. On nasa, where the zero-cost REPLACEALL reference already reaches a Tol_{rel} of 0.98, the separately trained zero-budget agent reaches 0.25, and on mercedes, where true values

TABLE A7
Full performance and wall-clock runtime. **DC** = DEMANDCLEAN. *Left*: 4 oracle-detection references (NF = NOFIX, DEL = DELETEALL, REPL = REPLACEALL, REPA = REPAIRALL, which equals the clean data) vs. **DC** (auto detection). *Middle*: GT% (including the calibration seed set), detector P_d/R_d, best external baseline (bold: exceeds the clean data), 1st/runner-up models with prior tolerance $\widehat{Tol}$ (Section IV-B1) and the score P of **DC** on the runner-up. *Right*: GT cells of the calibration seed set (Seed) and wall-clock seconds (**DC**: training, which includes detection, and inference, baselines: end-to-end). **Bold**/underline: best/runner-up among references+**DC**. C/R/CI marks the task type. Baseline abbreviations as in Table I.

	Oracle-Detection Ref.				Auto		Detection		vs. Baselines		1st Mod.		2nd Mod.				DC Time (s)			Baseline Time (s)			
Dataset	NF	DEL	REPL	REPA (clean data)	DC	GT%	P_d	R_d	Best Ext.		Mod.	$\widehat{Tol}$	Mod.	P	$\widehat{Tol}$	Seed	Train	Inf	R+B	HC	UC	Lop
adult^{C}	.852	**.856**	**.856**	**.856**	.854	0.1%	.17	.30	AC	.854	RF	0.03	GB	.868	−0.01	900	13,178	59	1,564	46	33	598
beers^{C}	.315	.307	.330	.355	**.361**	1.8%	.98	.99	RetClean	.353	RF	0.09	GB	.307	0.18	200	92	11	1,825	148	59	44
breast_c.^{C}	.957	.971	.986	.986	**.993**	2.9%	.81	.76	AC	.986	LR	0.60	GB	.979	0.00	200	37	4	305	58	25	28
smartf.^{C}	.962	.846	.962	.966	**.969**	0.2%	.98	1.00	Learn2Clean	**.967**	RF	0.19	GB	.797	−0.02	760	1,732	75	54,922	11,119	4,827	392
bike^{R}	.859	.865	.998	**1.00**	**1.00**	1.0%	.97	.99	COMET	1.00	LR	1.00	LASSO	1.00	1.00	640	4,571	200	71,774	2,965	16,700	236
mercedes^{R}	.360	.359	.408	.394	**.428**	2.0%	1.00	1.00	DIANA	**.402**	RF	0.05	RIDGE	.358	0.09	7,540	5,628	136	37,966	9,771	28,667	1,842
nasa^{R}	−.001	.528	.523	.532	**.533**	6.5%	.95	1.00	RetClean	.515	LR	0.53	RF	.680	0.19	120	53	<1	382	26	92	16
soilm.^{R}	.856	.749	.920	**.958**	.940	2.9%	.62	.86	AC	.937	GB	0.60	RF	.934	0.49	2,560	316	26	1,116	552	40	202
har^{CI}	.500	.908	.702	**.910**	.899	0.1%	.86	.97	RetClean	.802	Agglo.	0.41	KMeans	.900	0.34	320	24,174	164	1,449	2,611	3,224	255

TABLE A8
GT% of every method that reads clean training values, per dataset.

Method	adult	beers	breast_c.	smartf.	bike	mercedes	nasa	soilm.	har	flights	soccer	hospital
DELETEALL/REPLACEALL/REPAIRALL	60.0	60.0	59.9	60.0	60.0	60.0	59.9	59.9	60.0	60.0	60.0	60.0
RAHA+BARAN	<0.1	0.8	2.9	0.1	0.1	0.5	1.3	2.9	<0.1	0.8	0.2	2.0
LOPSTER	0.1	1.7	2.9	0.2	1.0	2.0	6.5	2.9	0.1	0	0	0
ACTIVECLEAN	1.1	20.7	29.9	2.1	2.9	11.9	29.9	29.9	0.7	21.0	4.7	30.0
COMET	56.0	48.0	53.9	56.8	56.2	59.8	50.0	59.5	3.2	51.4	54.0	56.8
RetClean	0	27.0	29.9	30.0	30.0	30.0	29.9	29.9	30.0	30.0	30.0	30.0
DEMANDCLEAN	0.1	1.8	2.9	0.2	1.0	2.0	6.5	2.9	0.1	<0.1	0	0

TABLE A9
Normalized Shapley values of the written cells ($\hat{\phi}_{\mathcal{W}}$) and the deleted rows ($\hat{\phi}_{\mathcal{D}}$) in a two-player attribution that reads no clean value, the deletion share of Table V, and the number of written cells and deleted rows. **Bold**: $\hat{\phi} > 0.05$, gray: $\hat{\phi} < -0.05$.

	Two players		Table V	Counts	
Dataset	$\hat{\phi}_{\mathcal{W}}$	$\hat{\phi}_{\mathcal{D}}$	$\hat{\phi}_{\mathcal{D}}$	$\lvert\mathcal{W}\rvert$	$\lvert\mathcal{D}\rvert$
adult	−.13	**.87**	.87	2	919
beers	**1.00**	.00	.00	2,690	0
breast_c.	**.60**	**.40**	.37	148	102
smartf.	**.91**	**.09**	.06	1,589	2,735
bike	**.96**	−.04	−.04	24,927	376
mercedes	**1.00**	.00	.00	58,043	0
nasa	**1.00**	.00	.00	459	0
soilm.	**1.00**	.00	.00	19,314	0
har	.01	**.99**	.99	53	20,513
flights	**1.00**	.00	.00	1,353	0
soccer	**1.00**	.00	.00	901	0
hospital	–	–	–	1,074	143

TABLE A10
Organically dirty datasets with the LLM baseline [1], scored as in Table IV. DC is DEMANDCLEAN, Repa, NF and Del are REPAIRALL, NOFIX and DELETEALL of Table IV, Ext. is the best external cleaner, GT% the GT usage of DEMANDCLEAN, and FM is Claude Opus 4.8 with the input tokens of all its turns and the list-price cost of its input and output tokens [2]. **Bold**: best cleaner.

Dataset	DC	Repa	NF	Del	Ext.	GT%	FM	Cost
beers	**.361**	.355	.315	.307	.353	1.8	.338	316K / $1.77
flights	**1.00**	1.00	.941	.834	1.00	<0.1	.941	188K / $0.84
soccer	**1.00**	1.00	.902	.856	1.00	0	.902	5.06M / $5.45
hospital	**1.00**	1.00	1.00	.995	1.00	0	1.00	329K / $1.44

carry 0.98 of the gain, it falls below NOFIX. The drop comes from missing labels at inference and from the removal of the cost signal that separates GT-repair from VE-replace in Stage 2 during training.

Variant 3: DEMANDCLEAN-Arch (Dueling → plain DQN network). This variant replaces the whole network configuration of DEMANDCLEAN, the Dueling head with its two 128-unit hidden layers and soft target updates, by a plain 64-64-32 DQN whose target network copies the online weights every fifth episode. The replacement lowers the mean Tol_{rel} of Figure 5 from 1.22 to 0.98, with the largest losses on beers, mercedes and smartfactory.

Variant 4: DEMANDCLEAN-Stage (two-stage → flat single). Flattening the hierarchy puts GT-repair into one argmax with the three zero-cost operations. The mean falls to 0.93, with the largest losses on beers (0.47 against 1.16), nasa and smartfactory.

Network configuration (Variant 3). The Dueling network outputs $Q(s,a) = V(s) + A(s,a) - \frac{1}{|A|}\sum_{a'} A(s,a')$. The centered advantages sum to zero over the actions, so every stored transition of a stage updates the shared $V(s)$ whichever action it records, and on a candidate whose actions lead to nearly equal returns the estimate rests on $V(s)$. This is the regime of the many low-impact candidates of Section IV-B2. The plain variant is a 64-64-32 ReLU network with one output per action, each output is fitted only from

the transitions that record its action, and its target network copies the online weights every fifth episode.

Two-stage hierarchy (Variant 4). In DEMANDCLEAN, Stage 1 compares no-op, delete and entering Stage 2, and every transition updates it, so the value of entering aggregates the returns of both Stage-2 outcomes. Stage 2 compares only VE-replace and GT-repair and learns from the transitions on which a cell was resolved in place, so its comparison of the two operations is fitted on exactly the cells where they differ in cost and in the restored value, with the budget features s_8 and s_9 entering through the gate g_k and the cost term λc_k of the reward described above. The flat variant is one Dueling network with four outputs and one replay buffer, so GT-repair competes in a single argmax with the three zero-cost operations, and the four actions share one value estimate per state.

The same outcome flips sign within a task type. Among the classification tasks, true values lower the score on `adult` and raise it on every other dataset with a gain. Among the regression tasks, they carry almost the whole gain on `mercedes` and `nasa` but less than half on `bike`, where estimates carry more and deletion lowers the score slightly. On `mercedes` the gain over NOFIX comes from the 57K cells that receive their true value, while restoring all 181K flagged cells, as REPAIRALL does, contracts the variance and scores lower (Finding 1), so the policy leaves 68% of the flagged cells unchanged. These sign flips show that task type gives a prior tendency at best and cannot substitute for per-instance state decisions, the empirical form of Challenge 1.

Attribution without clean values. Table A9 repeats the attribution of Table V with two players that the output of each deployed run and the dirty training split define alone. The written cells $\mathcal{W}$ are the kept training cells whose output value differs from the dirty value, and the deleted rows $\mathcal{D}$ are the training rows absent from the output. Separating a cell written with its true value from one written with an estimate requires the clean value, so $\mathcal{T}$ and $\mathcal{E}$ of Table V merge into $\mathcal{W}$. Every coalition is scored as in Table V, the empty coalition gives NOFIX and the full coalition gives the score of the run on all twelve datasets, and the computation reads no clean training value. On every dataset with a gain the deletion share differs from that of Table V by at most 0.04, and deletion carries the gain on `har` and `adult` in both attributions.

Lessons. The experiments support four lessons. First, the clean data is not the ceiling of cleaning, since DEMANDCLEAN exceeds it on five benchmark datasets. Second, the value of writing a true value depends on the dataset and changes sign within one task type, so a cleaner has to decide per cell whether a label pays off. Third, cost-free operations carry much of the gain, since value estimation writes more true values than the labeling budget pays for and deletion alone carries almost all of it on `har`. Fourth, decisions over the quality state transfer to new datasets of the same task type without retraining.

G FURTHER RELATED WORK

Studies of cleaning for machine learning measure how data errors and cleaning methods affect downstream models. CleanML [3] compares cleaning methods for five error types on classification tasks, Mohammed et al. [4] relate six data-quality dimensions to classification, regression and clustering performance, Neutatz et al. [5] examine when AutoML selects cleaning steps, and DIANA [6] turns such evidence into recommendations. These studies establish that data-level correctness and downstream gain diverge, which DEMANDCLEAN turns into the objective of a per-cell decision problem under a labeling budget.

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